

DECK MAP — HIDDEN (do not present)



- TL2 — Unity / MVP. Office Theme master, 11 layouts. Color/logo added by BC.
- FIXED FIRST LECTURE: 1 Title · 2 Introductions · T1 MVP (multi) · T2 Background+Camera (+copyright) · T3 Foreground art+walk anim · T4 Animated NPC · T5 Sound
- WAYPOINT / TL2 TAKE-OVER DIVIDER
- CORE vs OPTIONAL divider (ask BC for this semester's set)
- CORE: T6 FSM multi-state anim · T7 Collectibles (coins + poison=lose life) · T8 Power-ups (speed boost) · T9 HUD (coins+lives) · T10 Menu/Pause/Game-state (game-over at 0 lives)
- OPTIONAL: Moving platforms & hazards (rec. first) · Checkpoints/respawn (pairs w/ hazards) · Second enemy type · Particle effects/juice
- NOTE MODES: spoken word-for-word on intro/teaching slides; 'must do / how to research' brief on near-blank tutorial slides.
- SELF-DOCUMENTING: introduced here (HL2 Ravenholm), deep-dive TBD in a lighter TL3-TL6 slot; examinable for all.



TL2 — Unity & the MVP

Building a working platformer, live

Meet Your TL2 Team



- We are your TL2 group — everyone across the class who shares TL number 2 or 2+.
- Each of us will introduce ourselves by name.
 - Blake Hardy, Nick Howard, Jordan Hardy
- While one of us presents, the rest are circulating to help you.
- Ask us anything — we genuinely want the questions. (Don't forget to fill in the REC-LEC report when you do).

Topic 1 — The MVP: Why 'Playable' Comes First



- Even huge, multi-year projects aim for something playable early.
- If you don't have something that looks like a game in a few weeks, the project is in trouble.
- This ties directly to Continuous Delivery: always have a working build.
- Our MVP is grey-box: rectangles and a tic-tac player. It's boring — but it's a real, winnable game. (Pong has no graphics either.)

Topic 1 — Design Goal: The Level Teaches Itself



- Great games teach the player without words — through design alone.
- Much of what you've already built is quietly self-documenting: the Doppler siren says, "something's coming," parallax says "there's depth," your character's animation says "here's my direction and state" - all with no text.
- Half-Life 2, Ravenholm: a saw blade in a corpse + a blocked door teaches "saw blades are weapons" in ~10 seconds, no text. That's the same idea, turned up.
- Platformer: recognize the self-documentation already in your feature (practice - marked here, a safe place to learn it).
- Main project: your prefab must teach the player deliberately, saw-blade style. Start designing it that way as you write your Champion - you'll defend it at the Oral Exam.
- (A later TL group covers this in depth.)

Topic 1 — TUTORIAL: Build the grey-box MVP



Topic 2 — Background, Parallax & Camera



- An infinite auto-panelling sky: walk right forever (wall on the far left).
- 7 parallax layers for depth: sky, sun/moon, distant trees, closer buildings, platforms+character (suggest more).
- This presenter also sets up the camera to follow the player.
- The win flag sits on panel 3 — reachable, but you can stroll past it if you don't care to win.

Topic 2 — Using Others' Images:

Fair Use & the Oral Exam



- You'll use copyrighted images. Record the SOURCE and LICENSE of every asset — every time.
- At the Oral Exam, imagine you are selling this game for profit:
 - 1) Prove that you COULD be sued for copyright infringement.
 - 2) Then argue fair use in court — WITHOUT the 'it's educational' argument (you're selling it).
- You can only make that argument if you know exactly what you used and where it came from.

Topic 2 — TUTORIAL: Parallax background & camera follow



Topic 3:

Sprite & Walk Cycles



- Finding assets/sprites
 - (Unity Asset Store)
- Using sprite renderer
 - How to slice a sheet
- Unity animator & configuration
 - With a dash of scripting



Topic 3 TUTORIAL:

Sprites & Walk Cycles



Topic 4:

An Animated NPC



- Another game object
 - Own rigidbody, etc
- With its own walk cycle
 - And preset path
- Little easier from Topic 2



Topic 4 TUTORIAL: NPCs

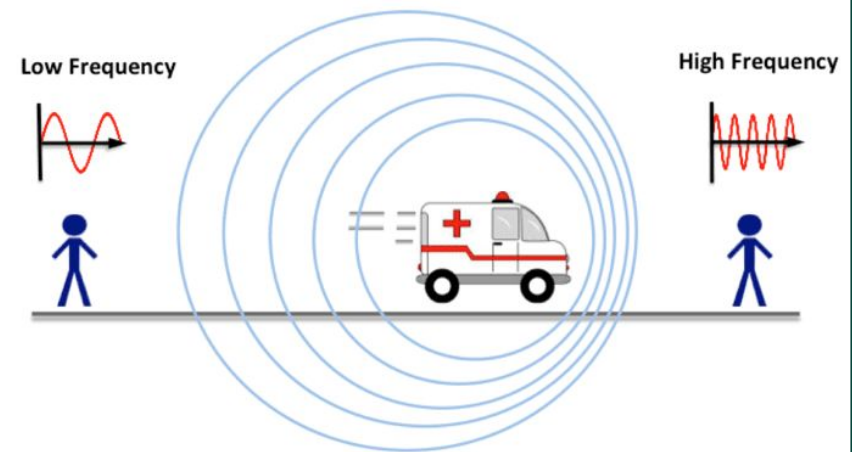


Topic 5:

Sounds & Doppler Effect



- Sound effects for collisions
 - More free assets
- NPC doppler effect
 - Distance -> volume
 - Motion -> pitch
- Unity makes this stupid easy



Topic 5 TUTORIAL: Sound Effects



What to hand in



- On Canvas each person submit your code and a video of your game running to:
Platformer Part 1 — Individual game code



Everything past here is a self-contained, independently-mergeable feature.

WAYPOINT — TL2 TAKES OVER HERE



CORE (must teach): FSM anim · Collectibles(coins+poison) · Power-ups · HUD · Menu/Pause/Game-state. OPTIONAL (as presenters allow): Moving platforms & hazards · Checkpoints · 2nd enemy · Particles. Ask BC to confirm this semester's set.

CORE FEATURES (EVERYONE) + OPTIONAL MODULES

Topic 6 — FSM Multi-State Character Animation [CORE]

- Give the character multiple animation states: at least idle, walk, jump, fall.
- Unity has a built in animator which is a finite state machine. You manage transitions between states (idle state -> jump state, for instance)
- Transitions between states are driven by conditions you control.



Topic 6 — TUTORIAL: Animator states & transitions [CORE]



Topic 7 — Collectibles: Coins & Poison [CORE]



- We want “coins” in our game. These are “good” items which raise our score.
- We also want at least one “poison” item that takes away a life.
- We will create a HUD which displays these later.
- This is an opportunity to show rather than tell. How can we make sure people know the “poison” item is bad?
- The game will end when lives reach 0 (after we implement it in Topic 10).



Topic 7 — TUTORIAL: Coin collection & the poison item [CORE]



Topic 8 — Power-ups: Speed Boost

[CORE]



- we want a “power -up” in our game - an item that helps the player instead of hurting them
- The SpeedBoost tmp multiplies the player’s move speed for a set duration
- It uses a trigger collider, so it activates on overlap rather than a physical collision
- This introduced a third kind of item beyond “good” and “bad”



Topic 8 — TUTORIAL: Speed-boost power-up [CORE]



Topic 9 — HUD: Score & Lives

[CORE]



- We want a HUD that shows the player's current lives and coin count at all times
- Lives are shown as a row of heart icons, each swapping between a full and empty sprite.
- A score readout updates live as coins are collected
- This is where scattered systems - coins, poison, power-ups become one readable status view

Topic 9 — TUTORIAL: Build the HUD [CORE]



Topic 10 — Main Menu, Pause & Game-State Flow [CORE]



- The full loop: Title/Menu -> Play -> Pause -> Win/Lose -> back to Menu.
- WIN: reach the flag. LOSE: lives hit zero
- Our final project needs this — that's why we learn it here.
- Reload scene to “reset game”

Topic 10 — TUTORIAL: Menu / pause / win / lose flow [CORE]



Topic 11 — Moving Platforms & Hazards [OPTIONAL — recommended first]



- Platforms that move on a path; a hazard (spikes / saw-blade-style obstacle).
- New technical idea: 'parent' the player to a moving platform so they ride it (classic Unity gotcha).
- Strong self-documenting feeder: a hazard's placement teaches danger wordlessly.
- If lethal, it becomes another game-over trigger feeding Topic 10.

Topic 11 — TUTORIAL: Moving platforms & a hazard [OPTIONAL]



Topic 12 — Checkpoints & Respawn [OPTIONAL — pairs with hazards]



- Respawn the player at the last checkpoint after a death.
- Teaches game-state persistence and 'save point' design.
- Most meaningful if a lethal hazard exists (Topic 11).

Topic 12 — TUTORIAL: Checkpoints & respawn [OPTIONAL]



Topic 13 — A Second Enemy Type

[OPTIONAL]



- A second enemy with different behavior — e.g. a ground patroller that turns at ledges.
- Teaches edge/ledge detection (how does it know not to walk off?).
- Reinforces the animation + AI lessons from Topic 4.

Topic 13 — TUTORIAL: Second enemy with edge detection [OPTIONAL]



Topic 14 — Particle Effects & Juice

[OPTIONAL]



- 'Juice': coin-collect sparkle, jump dust, win-screen confetti.
- Teaches Unity's particle system and why feedback makes a game feel good.
- Fully self-contained polish — great for an extra presenter.

Topic 14 — TUTORIAL: Particle effects [OPTIONAL]





What to hand in

- On Canvas each only 1 person from each TL group uploads the code and a video of the game running to:
Platformer Part 2b — Individual feature addition
- Each person add a short note saying which feature you added to the game.